OXFORD
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INTERNATIONAL GCSE
BIOLOGY
9201/2

Paper 2

Mark Scheme

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Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from aqa.org.uk

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer

with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	receptor cells		1	AO1 3.4.2b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.2	the computer used		1	AO4 3.4.1 6.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.3	$(480 + 505 + 495 + 500 + 470 =)$ 2450 $(2450 \div 5) = 490$		1 1	AO2 3.4.1 6.3(7)

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.4	did not discard / ignore the anomaly / error	allow they should not include 633 **or** they should not include the fourth attempt	1	AO4 3.4.1 6.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.5	to identify any anomalies to increase repeatability		1 1	AO4 3.4.1 6.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.6	any **one** from: • times are very / too short • includes the reaction time of the person pressing the stopwatch • resolution not good enough	allow descriptions such as delay before pressing button	1	AO4 3.4.1 6.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.7	any **one** from: • fast • automatic • protect from harm	allow (they are) not voluntary **or** does not involve (the conscious part of) the brain / thinking / decision **or** is not a conscious action	1	AO1 3.4.1c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.8	blinking when an insect flies towards the eye removing a hand from a hot object		1 1	AO2 3.4.1d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.9	gland muscle		1 1	AO1 3.4.1e

Total			**13**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.1	simple		1	AO1 3.6

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.2	extinct		1	AO1 3.6

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.3	hyracotherium		1	AO3 3.6

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.4	any **two** from: (Pliohippus) • taller (animal / horse) • longer bones in foot **or** fewer bones in foot **or** fewer toes • larger / longer / wider (back) teeth • Pliohippus existed after Mesohippus	answers must be comparative allow convese if clearly describing Mesohippus allow comparative use of figures from **Table 2**. allow has a hoof (instead of toes) allow has a narrower foot	2	AO3 3.6

02.5	**Level 3:** Relevant points (reasons/causes) are identified, given in detail and logically linked to form a clear account.	5-6	AO3 3.6.2b
	Level 2: Relevant points (reasons/causes) are identified, and there are attempts at logically linking. The resulting account is not fully clear.	3-4	AO3 3.6.2b
	Level 1: Points are identified and stated simply, but their relevance is not clear and there is no attempt at logical linking.	1-2	AO2 3.6.2b
	No relevant content	0	
	Indicative content *habitat drier or ground harder:* • feet not adapted to run on dry / hard ground **or** shorter horse not able to run fast on dry / hard ground • fewer trees to hide • (so) less able to escape from predators • reduced growth of plants for food *diet became tough or diet changed:* • smaller teeth not adapted to eat (tough) grass • lack of fruit and nuts in drier climate • (so) more difficult to obtain food *outcome:* • (therefore) less able to compete with others (better adapted) • (therefore) less likely to survive • (therefore) less likely to breed / reproduce • (therefore) genes / alleles not passed onto offspring For **Level 3** responses should consider both habitat and diet linked to outcome		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.6	expose horse to a noise	allow description	1	AO2 3.4.6cd
	reward calm behaviour (with praise / food)	ignore punish unwanted behaviour	1	
	keep exposing to even noisier situations **or** repeat until horse always stays calm in noisy conditions		1	
		if no other marks awarded allow **1** mark for operant conditioning		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.7	any **two** from: • greater visibility so people can find police • greater visibility to act as a deterrent • people are more likely to disperse / move out of the way • police / horses can move faster (in an emergency) • police are less likely to get injured	allow greater visibility so police can find people	2	AO3 3.4.6d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.8	sniffer dogs	allow hearing dogs for the deaf **or** guide dogs for the blind **or** search and rescue dog / horse **or** dogs to herd sheep allow other correct examples involving training such as mice to find land mines or guard dogs	1	AO1 3.4.6d

Total			**17**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.1	join / fuse / combine	allow fertilise	1	AO1 3.5.1a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.2	put a (plastic) bag around flower B		1	AO3 3.5.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.3	removed the male part(s) / male gametes (from flower B)	allow removed the pollen / anthers (from flower B)	1	AO3 3.5.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.4	3-layered triangular pyramid all three labels in food chain order	allow 2 marks for inverted pyramid correctly labelled	1 1	AO2 3.3.1c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.5	aphids reduce the amount of <u>wheat</u> produced	allow aphids eat the <u>wheat</u> allow so there is more <u>wheat</u> ignore cost	1	AO2 3.3.2a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.6	nucleus / chromosome	ignore DNA	1	AO1 3.5.2a 3.5.5b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.7	(an) enzyme	allow restriction enzyme	1	AO1 3.5.5b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.8	reduced pollution of land / water / environment (from toxic chemical) prevents reduction in biodiversity **or** no organisms / insects are killed	allow it may build up in the food chain ignore pollution unqualified allow so pesticide resistant aphid strains do not emerge ignore cost	1 1	AO3 3.3.4 3.5.5d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.9	any **two** from: • unknown effect on human health of eating the GM wheat • concern over possible spread of genes to other plants • aphid will affect other crops (if it no longer feeds off the wheat crop) • decrease in aphid population may affect other populations (in food web)	ignore ethical or religious reasons	2	AO2 3.5.5f

Total			**12**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.1	(heat the agar) to sterilise **or** to kill all bacteria in agar (press fingertips) to transfer bacteria (from the hands onto the agar) (secure the lid) to prevent contamination (from the environment) **or** to prevent bacteria from plate spreading (into the environment) (incubate) to give time for bacteria to reproduce / grow / form colonies **or** to give correct temperature for bacteria to reproduce / grow / form colonies	 allow description such to stop other bacteria getting in ignore culture bacteria	1 1 1 1	AO4 3.4.7 6.2 RP5

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.2	$193 - 34 = 159$ $\frac{159}{193}$ (×100) $= 82.38 / 82.4$ (%)	 allow 82%	1 1 1	AO2 3.4.7 6.3 (3)

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.3	any **three** from: • pick up disc using (sterile) forceps / tweezers • forceps / tweezers held in Bunsen flame **or** forceps / tweezers soaked in disinfectant / ethanol • lift the lid of the agar plate at an angle **or** do not fully open plate • work near a flame **or** work in a fume cupboard • replace the lid of the agar plate (whilst the next disc is collected)		3	AO4 3.4.7 6.2 RP5

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.4	use a ruler to measure the diameter (of the clear zone)		1	AO4 3.4.7 6.2 RP5
	divide by 2 **or** determine the radius	allow **2** marks for $a = \pi\ (d/2)^2$	1	AO2
	use πr^2 (to calculate the area)		1	AO2 6.3(14)

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.5	sanitiser diffused from discs		1	AO2 3.4.7f 3.1.5a
	(and) killed bacteria	allow prevented bacteria growing / reproducing	1	
	until <u>concentration</u> (of sanitiser) was too low to be effective		1	
		ignore zone of inhibition unqualified		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.6		do **not** accept immune bacteria in either mark point		AO2 AO1 3.4.7h
	non-resistant bacteria killed by (complex antimicrobial) chemical **or** resistant bacteria able to reproduce / replicate		1	
	(so) resistant bacteria out-compete non-resistant bacteria		1	
		allow resistant bacteria reproduce and pass on gene for resistance for **2** marks		

Total			**18**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.1	carbon dioxide increases (because) trees are burnt **or** (because) fewer trees absorb carbon dioxide **or** (because) fewer trees are photosynthesising **or** increased decay of tree debris	allow more heavy machinery / lorries (to remove logs) allow less carbon dioxide absorbed allow less photosynthesis ignore oxygen ignore habitats / biodiversity	1 1	AO2 3.3.3e 3.2.1b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.2	any **one** from: • less carbon dioxide released (per kg burned) • less carbon dioxide reduces global warming	allow biofuels are sustainable / renewable / carbon neutral	1	AO3 3.3.4ad

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.3	any **two** from: • nitrogen oxide is greater (by 10 au) • (more) nitrogen oxide increases acid rain • (5MJ) less energy released (per kg burned)	to award **2** marks there must be use of comparative data from Table **4** allow (1.13) times more biofuel needs to be burned (to get the same amount of energy) **or** less energy released (per kg burned) means more biofuel needs to be burned	2	AO3 3.3.4ab 6.3(12)

05.4	**Level 2:** A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.	3-4	AO3 3.3.4d
	Level 1: A conclusion, supported by some relevant reasons is given.	1-2	AO3 3.3.4d
	No relevant content	0	
	Indicative content **for the theory:** • overall increased carbon dioxide parallels overall increased temperature **or** positive correlation between concentration of carbon dioxide and temperature of air • use of data e.g. when carbon dioxide rose by 48 (au from 1992 to 2016) there was an increase of 0.75 (°C) in air temperature **against the theory:** • in some years e.g. 1995 – 1996 / 1999 – 2000 temperature falls while carbon dioxide is rising • many (large and small) rises and falls in temperature • other (unknown) factors may be involved in temperature change • (data may have been manipulated by) carefully chosen scales so they look like they coincide • overall correlation does not necessarily mean a causal link **or** not enough evidence to prove a link For **Level 2** there must be evidence for and against the statement including the use of some data from **Figure 5**		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.5	sea levels rising because polar ice caps melting **or** adverse weather patterns		1	AO2 3.3.4d
	(which causes) loss of homes / habitat / farmland	allow homes / farmland flooded	1	AO3
	crop failure due to drought / floods **or** food insecurity due to droughts / floods	allow crop failure due to (large scale) fires **or** loss of crops due to increase in crop-destroying insects **or** decrease in pollinators	1	AO2
	(which) could lead to starvation / famine		1	AO3
		allow for 2 marks increased number of pathogens (so) increased incidence of disease of people / crops allow for 2 marks large areas of land becoming uninhabitable so wars over what land remains if no other mark awarded allow sea levels rise and flood land / countries / areas / cities		

Total			**13**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.1	290÷85 3.4117.. 3.4	 allow correct significant figures from incorrect calculation using data from **Figure 6**	1 1 1	AO2 3.3.4b 6.3(1,4)

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.2	(excessive) fertiliser / nitrate ions added to crops (by farmer) nitrates washed / leached into river		1 1	AO2 3.3.4b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.3	(nitrate ions) cause the growth / increase of algae / plants (algae / plants) block light needed for photosynthesis (which) causes algae / plants to die microorganisms / bacteria cause decay of (these) dead algae / plants (and these microorganisms / bacteria) respire using up oxygen in the water (leading to the death of fish)	 allow light is blocked so less / no photosynthesis allow microorganisms / bacteria feed on (these) dead algae / plants allow reference to aerobic respiration do **not** allow reference to anaerobic respiration	1 1 1 1 1	AO2 AO1 3.3.3e 3.3.4b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.4	reduced competition from weeds **or** crops get more light / space / water (so) more photosynthesis / glucose (for a higher yield)	allow ions / minerals ignore nutrients do **not** accept oxygen	1 1	AO2 AO1 3.2.1b 3.3.2b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.5	(4% loss = 1.5 (kg per m^2) = 96% (yield) 1.5 ÷ 96 ×100 = 1.56(25 kg per m^2) (40 weeds gives 52% loss =) 48% (yield) (yield for 40 weeds per m^2 =) 1.56(25) × 48 ÷ 100 = 0.75 / 0.7488 / 0.749 (kg per m^2) **OR** 4% loss = 5 weeds (1) 4 ÷ 100 x 1.5 = 0.06 (1) (so maximum yield would be) 1.5 + 0.06 = 1.56 (1) (yield for 40 weeds per m^2) = 1.56 x 48 divided by 100 (1) = 0.75 / 0.7488 / 0.749 (kg per m^2) (1)	if (40 weeds gives 52% loss =) 48% (yield) is given as part of other incorrect working award **1** mark only	1 1 1 1 1	AO2 3.3.2b 6.3(3,11)

Total			**17**